CSE 219 — Signals and Linear Systems

Lec 3–4 · LTI Systems, Convolution, Causality and Stability

Lesson notes · Exam approach · TF practice with solutions · Quick notes (Anik Sir's portion)

How to Read These Notes

EXAM FOCUS	Exam importance, where it appears in the TF collection
MEMORIZE	Rules, tables and formulas to memorize
TRAP	Common mistakes and exam traps
EXAM-READY ANSWER	Model wording you can write in the exam
NOTE	Extra explanation / clarification
Grey box, monospace	Code, worked tables, derivations and hand-drawn sketches
Centered grey box	Formula to remember / final numerical result
Purple heading / ✓ green	Practice questions / their solutions

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Lesson 6 — Systems and Their Properties — linearity, time invariance (+ memory, causality, stability, invertibility)
Lesson 7 — Discrete Convolution as Polynomial Multiplication — reflect, shift, multiply, add
Lesson 8 — Impulse Response and the Convolution Sum / Integral — why y = x * h for every LTI system
Lesson 9 — Convolution Practice: DT, CT, Graphical, Interconnections — interval tables, cascades, parallel paths
Lesson 10 — Causality and Stability from h(t); h(t) of Simple Systems — h = 0 for t < 0; ∫|h| < ∞
Quick Notes — LTI and Convolution — merged revision notes + key diagrams

NOTE Slide numbers refer to Lecture-3-Linear-Time-Invariant-Systems-and-Convolution.pdf (59 slides, “Lec 3”) and Lecture-4-Convolution-Causality-and-Stability.pdf (70 slides, “Lec 4”). Memory, invertibility, interconnections and impulse responses of differential equations/circuits are textbook background (marked) — the EEE papers ask them every year.

Exam Map — What the TF Papers Ask

Convolution is the single most-asked Anik-part topic outside the transforms: 20 marks in CSE 219 2022-23 (Q8(a)), 15 in 2023-24 (Q1(b)), and a graphical convolution in nearly every EEE paper.

Session (paper)	Questions answered by this note
2023-24 (CSE 219)	Q1(b) $h_1=\delta[n]-\delta[n-2]$, $h_2=u[n-1]$, $h_3=\delta[n]-\delta[n-1]$; $(h_1+h_2)*h_3$: h_eq and y for x=u[n] (15) · Q3(b) Causal? stable? $e^{-4t}u(3-t)$, $e^{-6\lvert t\rvert}$, $te^{-t}u(t)$ (15)
2023-24 (EEE 211)	Q1(b) $dy/dt + t^2y = (2t+3)x$: linear? time-invariant? (7) · Q1(c) Interconnection of $h_1 \ldots h_5$ (exponentials, δ): BIBO? y for $x = e^{-t}u(t)$ (17) · Q2(a) Graphical convolution: x = rect on (1,2), h = triangle on (−1,1) (18) · Q2(b) Impulse response of a circuit in the time domain (17)
2022-23 (CSE 219)	Q8(a) Convolve $x[n] = 3^n u[-n-1] + (1/3)^n u[n]$ with $h[n] = (1/4)^n u[n+3]$ (20) · Q8(b) Causal? stable? $h_1 = e^{2t}u(-2-t)$, $h_2 = te^{-t}u(t)$ (10)
2022-23 (EEE 211)	Q5(b) LTI: x (V-shape) → y = rect; outputs for x(t−½)−x(t+½) and dx/dt (15) · Q7(b) x(t) = h(t) = t[u(t)−u(t−1)]: y(t) for all t (15) · Q8(b) $dy/dt + t^2y = (2t+3)x$: linear? time-invariant? (15)
2021-22 (EEE 211)	Q2(a) Graphical convolution (triangle x, rectangle-plus-impulse h); analytic y (17) · Q2(b) $y_1 = x_1h$, $y_2 = x_2h$ (triangle, trapezoid) [rest of wording cut off — illegible] (18)
2020-21 (EEE 211)	Q6(a) y = t·x(t+2): linear, causal, time-variant, memoryless, stable? (15) · Q6(b) Graphical convolution x = rect(t/2), h = r(t)u(1−t) (10) · Q6(c) Series RC, output v_R: impulse response by time-domain analysis (10)
2019-20 (EEE 211)	Q1 $y = (1/3)\int x(\tau)[\delta(t-\tau)+\delta(t-\tau+t_0)+\delta(t-\tau-t_0)]d\tau$: what it does; BIBO, linear, memoryless, causal, TI (35) · Q2 x = rect (0,2), h(t+1) = triangle: causal? y = x*h; what system does; z = x−x(t−4) (35)
2018-19 (EEE 211)	Q7(a) Linear system: x (rect) → y: causal, TI, memoryless? output for u(t)+u(t−1)−2u(t−2) (20) · Q7(b) h = u(t)−u(t−1), $x = t^2[u(t)-u(t-3)]$: graphical convolution (15)
2018-19 (CSE 311)	Q4(b) Prove y = x*h for an LTI system using time-domain arguments only (17)
2017-18 (EEE 211)	Q6(a) Response y_1 to u(t)−u(t−1) given; response to $x_2(t)$ (20) · Q6(b) $y = k_1\cos[k_2t + \int_0^t k_3x\,d\tau]$: TI, memoryless, causal, invertible, stable? (15) · Q7(a) Graphical convolution of x (Fig. 7a, 0–3) and h (−1 to 1) (20) · Q7(b) Cascade $h_1 = te^{-t}u$, $h_2 = e^{-2t}u$; x = u(t)−u(t−2)+δ(t+1): h, parallel equivalent, y (15)
2016-17 (EEE 211)	Q1(a) Memoryless/causal/stable: δ(t)−δ(t−2), 2rect(t/2), $2e^{-4t}u$, $(1-e^{-4t})u$ (15) · Q1(b) Interconnection $h_1 \ldots h_5$: overall h; y for x = u(t); stability (20) · Q3(c) RL circuit: impulse and step response (10)
2015-16 (EEE 301)	Q2(b) y = sin{x(t)+x(t−2)}: linear, TI, causal, memoryless, stable? (15) · Q3(b) $h = (e^{-3t}-e^{-2t})u$: zero-state response to u(t) and $e^{-t}u(t)$ (15)
2014-15 (EEE 301)	Q2(a) $y = \sin(2t)\log_{10}x(t)$; $y = e^{-j\omega t}\int x\, e^{-j\omega\tau}d\tau$: six properties (18) · Q2(b) y'+10y = 2x: h(t); output for x = Σ(−1)^k δ(t−5k) (17)
2014-15 (CSE 311)	Q7(a) Short notes: linear system, time-invariant system (10) · Q1(b) [L-3] Linearity and time invariance of an LTI system (10)
2013-14 (EEE 301)	Q2(b) $y = \int_0^t e^{-(t-\tau)}x(\tau)d\tau$: LTI? impulse response; BIBO (17)
2012-13 (EEE 301)	Q1(c) y = x_e − x_o: invertible? inverse (11) · Q2(a) $y = \int_{-\infty}^{2t-3}x\,d\tau$; $y = \tfrac{1}{2}\lvert x(t)\rvert - \pi t$: six properties (18) · Q2(b) δ-systems and rect(t/2): y for x = Σ(−1)^k δ(t−10k) (17)
2011-12 (EEE 301)	Q7(a) $y = 2x\cos\omega_c t$: linear, TI, causal, stable? (8) · Q7(c) Importance of y = x*h in real life (4) · Q7(d) $h_1 = \delta(t+4)$, $h_2 = \delta(t-4)$ summed, then triangle h_3; x = rect(−1,1): g, y (13)
2010-11 (EEE 301)	Q6(b)(i) Graphical convolution: x = 1 on (0,2), h = triangle on (−1,1) (10) · Q6(b)(ii) Interconnection $h_1 \ldots h_5$: impulse response; causal, stable? (15) · Q7(a) y = At·x(t): linear, TI, causal, stable? (10)

EXAM FOCUS Learn in this order: (1) graphical CT convolution with an interval table (EEE every year) · (2) DT convolution of one-sided exponentials split into left/right parts (CSE 219 2022-23 Q8(a), 20 marks) · (3) causal? stable? from h(t) (2022-23 Q8(b), 2023-24 Q3(b)) · (4) interconnected systems: h_eq then y (2023-24 Q1(b), EEE 2023-24 Q1(c)) · (5) test linear / time-invariant / causal / memoryless / stable for a given equation.

Lesson 6 — Systems and Their Properties

Slides 2–11 (Lec 3) · memory, causality, stability, invertibility: textbook

1. What a System Is (sl. 3–5)

A system takes a **whole input signal** and produces a whole output signal: x(t) → S → y(t) (or x[n] → y[n]). It is not a number-to-number function.

2. Linearity and Time Invariance (sl. 7–11)

- **Homogeneity**: a x(t) → a y(t). **Additivity**: $x_1 + x_2 \to y_1 + y_2$. Together: a x_1 + b x_2 → a y_1 + b y_2.
- **Time invariance**: $x(t - t_0) \to y(t - t_0)$: delaying the input only delays the output.
- **LTI** = both. Payoff: one special response (the impulse response) gives the output for every input.

3. Other Properties (textbook)

Property	Definition	Quick test
Memoryless	y(t) depends only on x(t) at the same t	any x(t − a), integral or derivative ⇒ memory
Causal	y(t) depends only on x(τ), τ ≤ t	look for x(t + a), x(−t), x(2t), upper limit > t
Stable (BIBO)	|x| ≤ B ⇒ |y| ≤ M	try x = constant, x = u(t); look for t·x, ∫x, 1/x, log x
Invertible	distinct inputs → distinct outputs	|x|, x^2, cos x, multiplying by something that can be 0 ⇒ not
Time-invariant	$x(t - t_0) \to y(t - t_0)$	t appearing outside x(·), or x(2t), x(−t) ⇒ variant

MEMORIZE Linearity test: (1) zero input must give zero output (an added constant or −πt breaks it); (2) y must contain x only to the first power, inside no non-linear function. Time-invariance test: compute $y_1 = S\{x(t - t_0)\}$ and $y(t - t_0)$ separately and compare.

TRAP • y = x(2t), y = x(−t) are linear but time-variant and non-causal. • A system with a coefficient depending on t (t^2y, t·x) is time-variant even though it is linear. • A differential equation is linear only with zero initial conditions.

Exam Approach

1. Linearity: apply a x_1 + b x_2 and compare with a y_1 + b y_2.
2. TI: compare $S\{x(t - t_0)\}$ with $y(t - t_0)$.
3. Causality/memory: list every time instant of x used to compute y(t).
4. Stability: find a bounded input with unbounded output (counterexample), or bound |y|.
5. Answer each property with one line of justification.

Practice — Lesson 6

(a) Actual — EEE 211 2019-20 Q1. (b) Actual — EEE 211 2023-24 Q1(b) / 2022-23 Q8(b). (c) Actual — EEE 211 2020-21 Q6(a). (d) Actual — EEE 211 2022-23 Q5(b). (e) Older papers, quick answers.

(a) $y(t) = (1/3)\int x(\tau)[\delta(t - \tau) + \delta(t - \tau + t_0) + \delta(t - \tau - t_0)]d\tau$. What does it do? Is it BIBO-stable, linear, memoryless, causal, time-invariant? (35)

(b) $dy/dt + t^2y = (2t + 3)x(t)$: linear? time-invariant?

(c) y(t) = t·x(t + 2): linear, causal, time-variant, memoryless, stable? (15)

(d) An LTI system maps x(t) = |t| (|t| ≤ 1, 0 elsewhere) to y(t) = 1 (|t| ≤ 1). Sketch the outputs for (i) x(t − ½) − x(t + ½), (ii) dx/dt. (15)

(e) (i) y = sin{x(t) + x(t − 2)}; (ii) y = sin(2t)$\log_{10}$x(t); (iii) y = $\int_{-\infty}^{2t-3}$x dτ; (iv) y = ½|x(t)| − πt; (v) y = x_e − x_o; (vi) y = 2x cos ω_c t; (vii) y = At·x(t); (viii) y = $\int_0^t e^{-(t-\tau)}x(\tau)d\tau$; (ix) y = $k_1\cos[k_2 t + \int_0^t k_3 x\, d\tau]$.

✓ Solution

(a) Sifting each impulse: y(t) = (1/3)[x(t) + x(t + t_0) + x(t − t_0)] — a **three-point moving average** (smoothing / low-pass filter). Linear: yes (sum of shifted copies). Time-invariant: yes (x(t − a) gives y(t − a)); h(t) = (1/3)[δ(t) + δ(t + t_0) + δ(t − t_0)]. Memoryless: no (uses t ± t_0). Causal: no for $t_0 \neq 0$ (uses x(t + t_0); h ≠ 0 at t = −t_0). BIBO-stable: yes, |y| ≤ (1/3)(B + B + B) = B (∫|h| = 1).

(b) Linear: if $x_1 \to y_1$, $x_2 \to y_2$ then a x_1 + b x_2 satisfies the equation with a y_1 + b y_2 (all terms are first power in x and y, zero initial state) ⇒ **linear**. Time-variant: with input x(t − t_0) the solution y_1 obeys y_1' + $t^2 y_1$ = (2t + 3)x(t − t_0), while y(t − t_0) obeys y′ + (t − t_0)²y = (2t − 2t_0 + 3)x(t − t_0) — different coefficients ⇒ **time-variant**.

(c) Linear (a x_1 + b x_2 → t(a x_1 + b x_2)(t + 2)). Non-causal (uses x(t + 2), the future). Time-variant: S{x(t − t_0)} = t x(t − t_0 + 2) ≠ y(t − t_0) = (t − t_0)x(t − t_0 + 2). Has memory (uses t + 2). Unstable: x = 1 gives y = t → ∞.

(d) Use linearity and time invariance on the given pair: (i) output = y(t − ½) − y(t + ½) = −1 on (−3/2, −½), 0 on (−½, ½), +1 on (½, 3/2). (ii) Differentiation commutes with an LTI system (derivative = limit of a difference of shifted inputs), so output = dy/dt = δ(t + 1) − δ(t − 1).

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(e)          linear  TI   causal  memoryless  stable  invertible
(i)           no     yes   yes       no         yes      no
(ii)          no     no    yes       yes        no       no (sin 2t = 0 at some t)
(iii)         yes    no    no        no         no       yes (y'(t) = 2x(2t - 3))
(iv)          no     no    yes       yes        no       no (|x|)
(v) = x(-t)   yes    no    no        no         yes      yes (inverse = itself)
(vi)          yes    no    yes       yes        yes      no (cos ω_c t = 0)
(vii)         yes    no    yes       yes        no       no (t = 0)
(viii)        yes    for inputs starting at t ≥ 0 it is LTI with h = e^{-t}u(t); ∫|h| = 1 ⇒
stable
(ix)          no     no    yes       no         yes (|y| ≤ |k₁|)
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Lesson 7 — Discrete Convolution as Polynomial Multiplication

Slides 12–31 (Lec 3)

Multiply A(z) = 1 + 3z + 2z² + 5z³ + 4z⁴ by B(z) = 2 − z + 4z² + z³ + 3z⁴. The coefficient of z⁴ collects every product whose powers add to 4 (sl. 14–15):

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c[4] = a0b4 + a1b3 + a2b2 + a3b1 + a4b0 = 1·3 + 3·1 + 2·4 + 5·(−1) + 4·2 = 17
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$$\mathbf{c[n] = (a * b)[n] = \Sigma_k\, a[k]\, b[n-k]}$$

b[n − k] means: **reflect** b (b[−k]), then **shift** it right by n, multiply the samples that line up, and add (sl. 17–22). Convolution is commutative: a * b = b * a (polynomial product order does not matter, sl. 29–30).

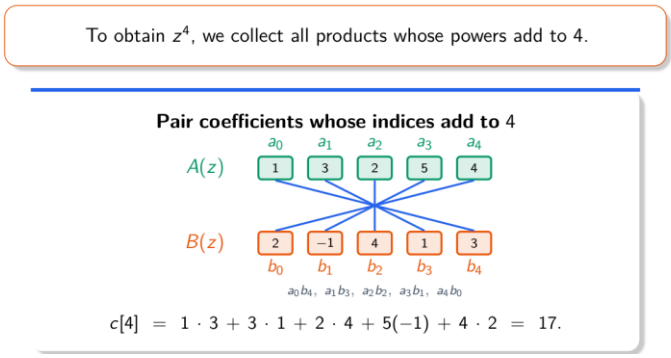


Figure 7.1 — The z⁴ coefficient comes from criss-crossed pairings (Lec 3, sl. 15)

Worked Example (sl. 23–28)

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x = {1, 2, 1, 1},  h = {1, 1, −1, 2}  (both start at n = 0)
n:     0    1    2    3    4    5    6
y:     1    3    2    2    4    1    2
e.g. y[3] = 1·2 + 2·(−1) + 1·1 + 1·1 = 2;   length = 4 + 4 − 1 = 7
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MEMORIZE Finite sequences: start index = sum of start indices, length = L + M − 1, Σy = (Σx)(Σh) (quick check: 5 × 3 = 15 = 1+3+2+2+4+1+2 ✓). Fast hand method: multiply as polynomials, or use a sliding-strip table.

Exam Approach

1. Write both sequences with their index ranges.
2. Reflect h, slide one step at a time, multiply-add the overlap (or multiply polynomials).
3. Check: start index, length, and Σy = Σx·Σh.

Practice — Lesson 7

(a) Practice set 3.1. (b) Practice set 3.3. (c) Constructed.

(a) x = {1, 2} (n = 0, 1), h = {1, −1, 2} (n = 0, 1, 2). Find y = x * h.

(b) x[n] is zero except x[−1] = 2, x[0] = −1, x[2] = 3. Write x as shifted impulses and y in terms of h.

(c) Find the coefficient of z⁵ in A(z)B(z) of the slide example.

✓ Solution

(a) $(1 + 2z)(1 - z + 2z^2) = 1 + z + 0z^2 + 4z^3 \rightarrow y = \{1, 1, 0, 4\}$, $n = 0 \ldots 3$. Check Σ: $3 \cdot 2 = 6$ ✓.

(b) $x[n] = 2\delta[n + 1] - \delta[n] + 3\delta[n - 2] \Rightarrow y[n] = 2h[n + 1] - h[n] + 3h[n - 2]$.

(c) $c[5] = a1b4 + a2b3 + a3b2 + a4b1 = 3 \cdot 3 + 2 \cdot 1 + 5 \cdot 4 + 4 \cdot (-1) = 27$.

Lesson 8 — Impulse Response and the Convolution Sum / Integral

Slides 32–57 (Lec 3), 2–9 (Lec 4)

Philosophy (sl. 34–37): solve for the simplest input, break any input into such pieces, combine the answers (like counting factors of 72 = $2^3 \cdot 3^2$: (3+1)(2+1) = 12).

- Simplest input: δ[n]. Its output is the **impulse response** h[n] — the system's signature (sl. 38–40).
- Echo system: h[n] = u[n] (equal echoes forever); h = α^nu[n], 0 < α < 1: fading echoes, larger α fades slower (sl. 42–44). Pure delay by 3: h[n] = δ[n − 3] (CT: δ(t − 3)) (sl. 45–46).
- Input x = δ[n] + 2δ[n−1] − δ[n−2] + δ[n−3] into h = u[n]: y = u[n] + 2u[n−1] − u[n−2] + u[n−3] = 1, 3, 2, 3, 3, … (sl. 47–51).

Derivation (sl. 52–57) — exam-ready

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1. Any input is a sum of shifted, scaled impulses:  x[n] = Σ_k x[k] δ[n − k]
2. Time invariance:  δ[n − k] → h[n − k]
3. Homogeneity:      x[k]δ[n − k] → x[k]h[n − k]
4. Additivity:       y[n] = Σ_k x[k] h[n − k] = (x * h)[n]
CT (textbook): x(t) = ∫ x(τ) δ(t − τ) dτ (sifting) → same steps with ∫ ⇒ y(t) = ∫ x(τ) h(t − τ) dτ
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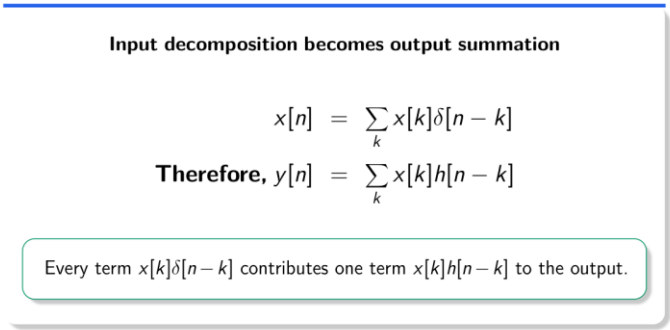


Figure 8.1 — Input decomposition becomes output summation (Lec 3, sl. 55)

CT convolution (Lec 4, sl. 6–8): y(t) = ∫ x(τ)h(t − τ)dτ. Fix t, slide h(t − τ) along τ, integrate the product over the overlap. Same recipe: choose the output point, reflect and shift, multiply and add.

EXAM-READY ANSWER Why convolution matters in real life (EEE 301 2011-12 Q7(c)): once h(t) is measured (hit the system with an impulse — clap in a hall, tap a structure), the output for every input is x * h without re-solving the system: reverb/echo simulation, sensor smoothing (moving average), image blurring and sharpening, channel modelling in communications.

Practice — Lesson 8

(a) Actual — CSE 311 2018-19 Q4(b). (b) Actual — EEE 301 2011-12 Q7(c). (c) Practice set 4.9–4.11.

(a) Prove that for an input x(t), the output of an LTI system with impulse response h(t) is y = x * h, using only time-domain arguments. (17)

(b) Mention the importance of y = x * h in real-life applications. (4)

(c) How can convolution simulate a large hall, smooth sensor readings, and add an echo?

✓ Solution

(a) By the sifting property $x(t) = \int x(\tau)\delta(t - \tau)d\tau$: x is a continuum of impulses of strength $x(\tau)d\tau$ at $t = \tau$. Let $\delta(t) \rightarrow h(t)$. Time invariance: $\delta(t - \tau) \rightarrow h(t - \tau)$. Homogeneity: $x(\tau)d\tau\ \delta(t - \tau) \rightarrow x(\tau)d\tau\ h(t - \tau)$. Additivity (the integral is a limit of sums): $y(t) = \int x(\tau)h(t - \tau)d\tau = x(t) * h(t)$. Every step uses only L and TI.

(b) See the EXAM-READY ANSWER above.

(c) Hall: record the hall's impulse response (a clap or balloon pop) and convolve the dry voice with it. Smoothing: convolve readings with a short averaging kernel $h = (1/M)\Sigma\delta[n - k]$, $k = 0 \ldots M-1$. Echo: $h[n] = \delta[n] + a\delta[n - D]$; D sets the delay, a (< 1) the strength; more taps ($a, a^2, \ldots$) give repeated echoes.

Lesson 9 — Convolution Practice: DT, CT, Graphical, Interconnections

Slides 10–52 (Lec 4) · interconnections: textbook

1. Slide Examples

Example	Signals	Result
Ex 2.2 (sl. 11–19)	x = {0.5, 2}, h = {1, 1, 1}	y = {0.5, 2.5, 2.5, 2}, n = 0…3
Ex 2.3 (sl. 20–27)	$\alpha^n u[n] * u[n]$	$(1 - \alpha^{n+1})/(1 - \alpha)\ u[n] \to 1/(1 - \alpha)$
Ex 2.5 (sl. 28–34)	$2^n u[-n] * u[n]$	2^{n+1} (n < 0); 2 (n ≥ 0)
Ex 2.6 (sl. 35–40)	$e^{-at}u(t) * u(t)$	$(1/a)(1 - e^{-at})u(t)$
Ex 2.10 (sl. 42–47)	$[(½)^n u[n] + 2^n u[-n]] * u[n]$	2^{n+1} (n < 0); $4 - 2^{-n}$ (n ≥ 0)
sl. 48–52	$e^t u(-t) * e^{-t}u(t)$	$½e^t$ (t < 0); $½e^{-t}$ (t ≥ 0) = $½e^{-\|t\|}$

Key step of every infinite-length example: find the overlap. For $x[k] = \alpha^k u[k]$ and $h[n - k] = u[n - k]$, x is nonzero for $k \geq 0$ and $h[n - k]$ for $k \leq n \Rightarrow$ overlap $0 \leq k \leq n$ (empty for $n < 0$). Then sum a geometric series:

$$\Sigma_{k=0}^{n} \alpha^k = (1 - \alpha^{n+1})/(1 - \alpha) \qquad \Sigma_{k=-\infty}^{m} \beta^k = \beta^m/(1 - 1/\beta)\ \ (\beta > 1)$$

Distributivity (sl. 41): $x * (h_1 + h_2) = x * h_1 + x * h_2$ — split a two-sided input into its left and right parts, convolve separately, add.

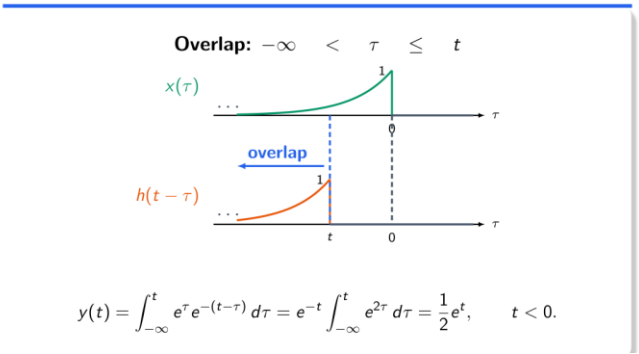


Figure 9.1 — CT overlap for t < 0 ends at τ = t (Lec 4, sl. 50)

2. Graphical CT Convolution — the Interval Table

- Choose the simpler signal to flip (often the rectangle).
- Write the edges of h(t − τ) as functions of t; list the t-intervals where the overlap changes (edge of one passes edge of the other).
- On each interval write the integral limits and evaluate; check continuity at interval ends.
- Useful shortcut: x = rect on (a, b) ⇒ $y(t) = \int_{t-b}^{t-a} h(s)ds = H(t - a) - H(t - b)$, H = running integral of h.

3. Interconnections (textbook)

Cascade: $h = h_1 * h_2$	Parallel: $h = h_1 + h_2$	$\delta(t - a) * h(t) = h(t - a)$

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Useful pair:  e^{-at}u(t) * e^{-bt}u(t) = [e^{-bt} - e^{-at}]/(a - b) · u(t)     (a ≠ b)
              e^{-at}u(t) * e^{-at}u(t) = t e^{-at}u(t)
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TRAP • Do not forget the n < 0 (t < 0) region: one-sided signals often give a nonzero left tail (Ex 2.5). • u[n + 3] starts at n = −3: shift the summation limits. • Interval ends are where the edges meet — write them as t-values before integrating.

Exam Approach

1. Sketch both signals; decide which to flip.
2. Find the overlap limits as functions of n (or t) and split into cases.
3. Evaluate each sum/integral; geometric series for exponentials.
4. Check continuity at case boundaries, the support (sum of supports) and the area ($\int y = \int x \cdot \int h$).

Practice — Lesson 9

(a) Actual — CSE 219 2022-23 Q8(a). (b) Actual — CSE 219 2023-24 Q1(b). (c) Actual — EEE 211 2023-24 Q1(c).

(a) Convolve $x[n] = 3^n u[-n-1] + (1/3)^n u[n]$ with $h[n] = (1/4)^n u[n+3]$. (20)

(b) $h_1 = \delta[n] - \delta[n-2]$ and $h_2 = u[n-1]$ in parallel, followed by $h_3 = \delta[n] - \delta[n-1]$. (i) h_eq; (ii) y for x = u[n]. (15)

(c) x feeds three paths — h_1; h_2 followed by h_3; h_4 — which are added and passed through h_5. $h_1 = e^{-2t}u$, $h_2 = e^{-2t}u$, $h_3 = e^{-t}u$, $h_4 = \delta(t)$, $h_5 = e^{-3t}u$. Using time-domain techniques: (i) is it BIBO stable? (ii) y for $x = e^{-t}u(t)$. (17)

✓ Solution

(a) Split $x = x_1 + x_2$ (distributivity), $x_1 = 3^n u[-n-1]$, $x_2 = (1/3)^n u[n]$. $h[n-k] = (¼)^{n-k}$ is nonzero for $k \le n+3$.

```
y1[n] = Σ_{k ≤ min(-1, n+3)} 3^k (1/4)^{n-k} = 4^{-n} Σ 12^k
  n ≤ -4: upper limit n+3:  4^{-n}·12^{n+3}·(12/11) = 256·3^{n+4}/11
  n ≥ -3: upper limit -1:   4^{-n}·(1/12)/(1 - 1/12) = 4^{-n}/11
y2[n] = Σ_{k=0}^{n+3} (1/3)^k (1/4)^{n-k}  (n ≥ -3) = 4^{-n} Σ_{k=0}^{n+3} (4/3)^k
      = 4^{-n}·[(4/3)^{n+4} - 1]/(1/3) = 256·3^{-(n+3)} - 3·4^{-n};   0 for n < -3

y[n] = 256·3^{n+4}/11                        n ≤ -4
       256·3^{-(n+3)} - (32/11)·4^{-n}        n ≥ -3
Values: y[-4] = 23.27, y[-3] = 69.82, y[-2] = 38.79, y[-1] = 16.81, y[0] = 6.57, y[1] = 2.43
```

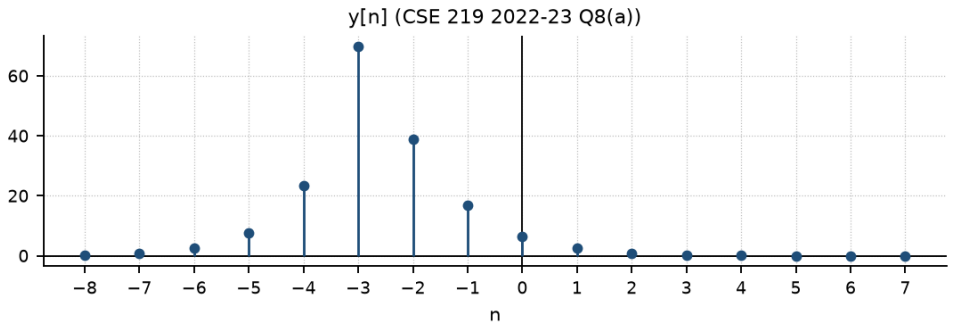


Figure 9.2 — Output of CSE 219 2022-23 Q8(a): rises as 3^n on the left, peaks at n = −3, decays on the right

(b) Parallel then cascade: $h_{eq} = (h_1 + h_2) * h_3$. $u[n-1] * (\delta[n] - \delta[n-1]) = u[n-1] - u[n-2] = \delta[n-1]$; $(\delta[n] - \delta[n-2]) * (\delta[n] - \delta[n-1]) = \delta[n] - \delta[n-1] - \delta[n-2] + \delta[n-3]$. Adding:

```
(i)  h_eq[n] = δ[n] - δ[n - 2] + δ[n - 3]
(ii) y[n] = u[n] * h_eq = u[n] - u[n - 2] + u[n - 3]
     n:   <0   0   1   2   3   4 …
     y:    0   1   1   0   1   1 …
```

(c) Middle path: $h_2 * h_3 = e^{-2t}u * e^{-t}u = (e^{-t} - e^{-2t})u$. Sum of the three paths: $g = e^{-2t}u + (e^{-t} - e^{-2t})u + \delta(t) = \delta(t) + e^{-t}u(t)$. Then

```
h_eq = g * h5 = e^{-3t}u + e^{-t}u * e^{-3t}u = e^{-3t}u + ½(e^{-t} − e^{-3t})u = ½(e^{-t} +
e^{-3t})u(t)
(i)  ∫|h_eq| dt = ½(1 + 1/3) = 2/3 < ∞  ⇒ BIBO stable (also causal)
(ii) y = e^{-t}u * ½(e^{-t} + e^{-3t})u = ½ t e^{-t}u + ¼(e^{-t} − e^{-3t})u
     y(t) = [ ½ t e^{-t} + ¼ e^{-t} − ¼ e^{-3t} ] u(t)
```

Practice — Lesson 9 (continued) — Graphical Convolution

(d) Actual — EEE 211 2023-24 Q2(a). (e) Actual — EEE 211 2019-20 Q2. (f) Actual — EEE 211 2021-22 Q2(a). (g) Actual — EEE 211 2020-21 Q6(b). (h) Actual — EEE 211 2022-23 Q7(b). (i) Older: EEE 301 2010-11 Q6(b)(i), EEE 211 2018-19 Q7(b), EEE 301 2011-12 Q7(d), 2012-13 Q2(b), 2015-16 Q3(b).

(d) x = 1 on (1, 2); h = triangle on (−1, 1), peak 1 at 0. Find y = x * h graphically. (18)

(e) x = 1 on (0, 2); h(t + 1) is the triangle on (−1, 1). (i) Causal? (ii) y = x * h. (iii) What does the system do? (iv) Output for z = x(t) − x(t − 4). (35)

(f) x = triangle on (0, 2), peak 1 at t = 1; h = 2 on (0, 1) plus an impulse δ(t − 3). Find y graphically and analytically. (17)

(g) x = rect(t/2), h = r(t)u(−t + 1). Find y. (10)

(h) x(t) = h(t) = t[u(t) − u(t − 1)]: find y(t) for all t. (15)

(i) Older papers (see solutions).

✓ Solution

(d) Flip the rectangle: $y(t) = \int_{t-2}^{t-1} \text{tri}(s)ds$. Let F(s) = running integral of the triangle: 0 (s < −1), $(s + 1)^2/2$ (−1..0), $1 - (1 - s)^2/2$ (0..1), 1 (s > 1). y = F(t − 1) − F(t − 2):

```
interval    overlap                   y(t)
t < 0       none                      0
0 ≤ t ≤ 1   left edge of tri          t²/2
1 ≤ t ≤ 2   rect covers the peak      −t² + 3t − 3/2      (max 3/4 at t = 1.5)
2 ≤ t ≤ 3   right edge                (3 − t)²/2
t > 3       none                      0           check: area = 1·1 = 1 ✓
```

(e) (i) h(t) = triangle on (0, 2) with peak 1 at t = 1 ⇒ h = 0 for t < 0 ⇒ **causal**. (ii) y = F(t − 1) − F(t − 3):

```
0 ≤ t ≤ 1: t²/2        1 ≤ t ≤ 2: 1 − (2 − t)²/2        2 ≤ t ≤ 3: 1 − (t − 2)²/2        3 ≤ t
≤ 4: (4 − t)²/2
(iii) a smoothing (low-pass, triangular-weighted moving average) filter with a delay of 1: the
sharp
      edges of x become smooth S-curves, the duration grows from 2 to 4, peak y(2) = 1, area
2·1 = 2 ✓
(iv) z → y(t) − y(t − 4): the pulse above on [0, 4] followed by its negative on [4, 8]
```

(f) h = 2[u(t) − u(t − 1)] + δ(t − 3). The impulse part gives x(t − 3) (triangle on (3, 5)). The rectangle part: $2\int_{t-1}^{t} x(s)ds$:

```
0 ≤ t ≤ 1: t²      1 ≤ t ≤ 2: −2t² + 6t − 3      2 ≤ t ≤ 3: (3 − t)²
plus 3 ≤ t ≤ 4: t − 3,  4 ≤ t ≤ 5: 5 − t   (the delayed triangle)
y(t) = t²[u(t) − u(t−1)] + (−2t² + 6t − 3)[u(t−1) − u(t−2)] + (3−t)²[u(t−2) − u(t−3)] + x(t −
3)
```

(g) x = 1 on (−1, 1); h = t on (0, 1). $y = \int_{t-1}^{t+1} h(s)ds = H(t + 1) - H(t - 1)$, $H(s) = s^2/2$ on [0,1], ½ after:

```
−1 ≤ t ≤ 0: (t + 1)²/2      0 ≤ t ≤ 1: 1/2      1 ≤ t ≤ 2: 1/2 − (t − 1)²/2      else 0
```

(h) Overlap $0 \le \tau \le t$ for $0 \le t \le 1$ and $t - 1 \le \tau \le 1$ for $1 \le t \le 2$:

```
0 ≤ t ≤ 1: ∫₀ᵗ τ(t − τ)dτ = t³/6
1 ≤ t ≤ 2: ∫_{t−1}^{1} τ(t − τ)dτ = −t³/6 + t − 2/3      (continuous: 1/6 at t = 1, 0 at t = 2)
otherwise 0
```

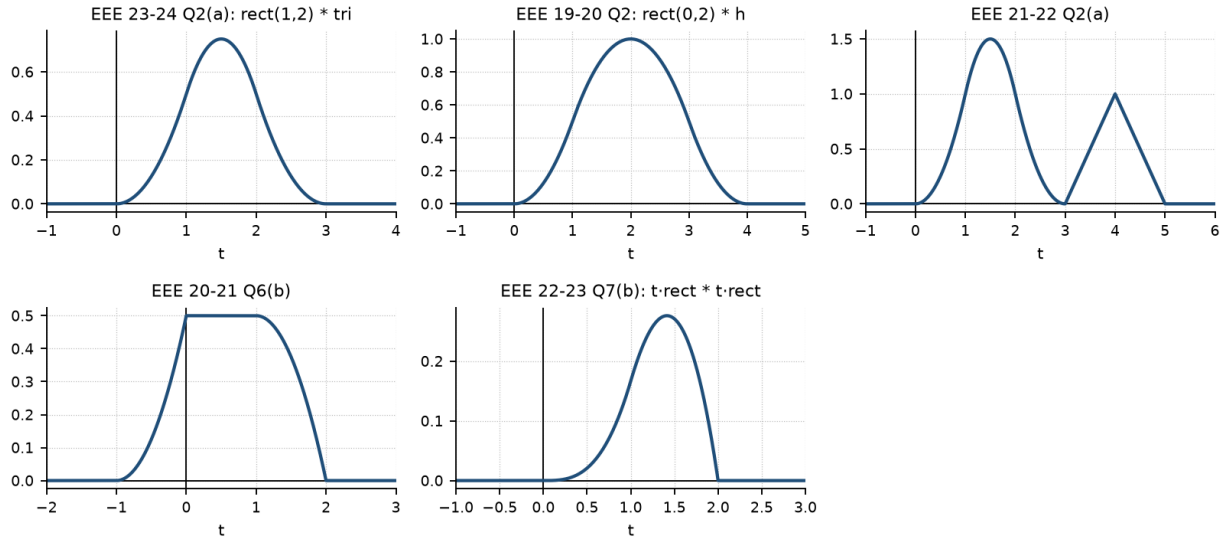


Figure 9.3 — Outputs of the graphical-convolution questions (d)–(h)

(i) EEE 301 2010-11 Q6(b)(i): x = 1 on (0, 2), h = tri: (t+1)²/2 [−1,0]; 1 − (1−t)²/2 [0,1]; 1 − (t−1)²/2 [1,2]; (3−t)²/2 [2,3]. EEE 211 2018-19 Q7(b): x = t² on (0, 3), h = 1 on (0, 1): y = t³/3 [0,1]; [t³ − (t−1)³]/3 [1,3]; [27 − (t−1)³]/3 [3,4]. EEE 301 2011-12 Q7(d): g = x(t + 4) + x(t − 4); y = P(t + 4) + P(t − 4) with P = rect(−1,1) * tri = (t+2)²/2 [−2,−1], 1 − t²/2 [−1,1], (2−t)²/2 [1,2]. EEE 301 2012-13 Q2(b): δ(t+1)*δ(t−1) = δ(t), so h = δ(t) + rect(t/2); y = Σ(−1)ᵏ[δ(t − 10k) + rect((t − 10k)/2)]. EEE 301 2015-16 Q3(b): step response −1/6 + ½e^{−2t} − ⅓e^{−3t}; response to e^{−t}u: −½e^{−t} + e^{−2t} − ½e^{−3t} (all × u(t)). EEE 211 2017-18 Q7(b): h_1 h_2 = (t − 1)e^{-t} + e^{-2t} = t e^{-t} − e^{-t} + e^{-2t} (three parallel branches).

Lesson 10 — Causality and Stability from h(t); h(t) of Simple Systems

Slides 53–69 (Lec 4) · impulse responses of differential equations and circuits: textbook

causal ⇔ h(t) = 0 for t < 0 (h[n] = 0, n < 0)
BIBO stable ⇔ ∫|h(t)|dt < ∞ (Σ|h[n]| < ∞)

Why (sl. 58): $|y(t)| = |\int h(\tau)x(t-\tau)d\tau| \le B\int|h(\tau)|d\tau$. Periodic h (sl. 60–62): a nonzero periodic h cannot be causal (any nonzero value repeats at $t_0 - kT < 0$) and cannot be stable (area A per period, infinitely many periods).

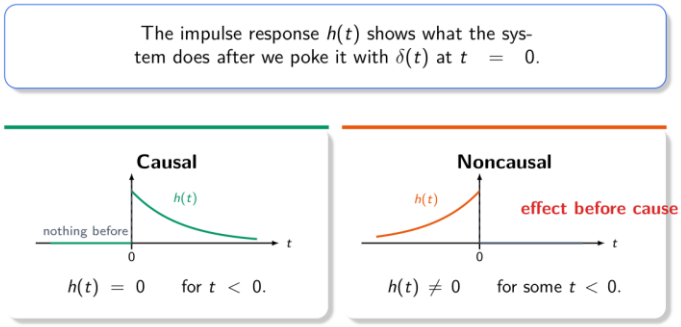


Figure 10.1 — Causal vs non-causal impulse responses (Lec 4, sl. 55)

```
Slide examples:  h1 = e^{2t}u(-2 - t): nonzero for t ≤ -2 ⇒ non-causal;  ∫_{-∞}^{-2} e^{2t} =
e^{-4}/2 ⇒ stable
                 h2 = t e^{-t}u(t): causal;  ∫₀^∞ t e^{-t} dt = 1 ⇒ stable
Minimum delay to make e^{-2t}u(t + 2) causal (sl. 68–69): start -2 + a ≥ 0 ⇒ a ≥ 2
```

Impulse Response of a First/Second-Order System (textbook)

```
y' + a y = b x          ⇒ h(t) = b e^{-at}u(t)
Series RC, output v_C: RC v' + v = x  ⇒ h_C = (1/RC) e^{-t/RC} u(t)
Series RC, output v_R = x - v_C       ⇒ h_R = δ(t) - (1/RC) e^{-t/RC} u(t)
Series L, shunt C (output v_C): LC y'' + y = x ⇒ h = ω₀ sin(ω₀t) u(t), ω₀ = 1/√(LC)
Method: for t > 0 solve the homogeneous equation; the impulse sets the initial conditions
        (order n: h^{(n-1)}(0⁺) = 1/a_n, lower derivatives 0; a δ in h appears if numerator
order = n)
```

TRAP • u(3 − t) and u(−2 − t) extend to −∞: non-causal. • $e^{-4t}u(3-t)$ grows as t → −∞: unstable. • Even $e^{-6|t|}$ is non-causal (nonzero for t < 0) though stable.

Exam Approach

1. Causality: find the support of h; any nonzero part at t < 0 ⇒ non-causal.
2. Stability: compute ∫|h| (split piecewise); exponentials: check decay in both directions.
3. For an equation/circuit: get the ODE, solve the homogeneous part for t > 0, impose the impulse jump conditions.

Practice — Lesson 10

(a) Actual — CSE 219 2022-23 Q8(b). (b) Actual — CSE 219 2023-24 Q3(b). (c) Actual — EEE 211 2023-24 Q2(b). (d) Actual — EEE 211 2020-21 Q6(c). (e) Older: EEE 211 2016-17 Q1(a), Q3(c); EEE 301 2014-15 Q2(b).

(a) Causal? stable? $h_1 = e^{2t}u(-2-t)$, $h_2 = te^{-t}u(t)$. (10)

(b) Causal and/or stable? (i) $e^{-4t}u(3-t)$; (ii) $e^{-6|t|}$; (iii) $te^{-t}u(t)$. (15)

(c) Find, in the time domain, the impulse response of the LC circuit (series L, output across shunt C). (17)

(d) RC series circuit, output v_R, initially relaxed: h(t) by time-domain analysis. (10)

(e) (i) memoryless/causal/stable: $\delta(t)-\delta(t-2)$, $2\text{rect}(t/2)$, $2e^{-4t}u$, $(1-e^{-4t})u$; (ii) $y' + 10y = 2x$: h(t) and output for $x = \Sigma_{k=-2}^{2}(-1)^k\delta(t-5k)$.

✓ Solution

(a) h_1: nonzero for $t \le -2 \Rightarrow$ **non-causal**; $\int_{-\infty}^{-2} e^{2t}dt = e^{-4}/2 < \infty \Rightarrow$ **stable**. h_2: zero for $t < 0 \Rightarrow$ **causal**; $\int_0^\infty te^{-t}dt = 1 \Rightarrow$ **stable**.

(b) (i) nonzero for all $t < 3 \Rightarrow$ non-causal; $\int_{-\infty}^{3} e^{-4t}dt = \infty$ (grows as $t \to -\infty$) $\Rightarrow$ unstable. (ii) nonzero for $t < 0 \Rightarrow$ non-causal; $\int e^{-6|t|} = 2\cdot(1/6) = 1/3 \Rightarrow$ stable. (iii) causal, $\int = 1 \Rightarrow$ stable.

(c) KVL: $x = L\,di/dt + y$, $i = C\,dy/dt \Rightarrow LC\,y'' + y = x$. For $t > 0$ with $x = \delta$: $LC\,h'' + h = 0 \Rightarrow h = A\cos\omega_0 t + B\sin\omega_0 t$, $\omega_0 = 1/\sqrt{(LC)}$. Integrating the equation over $(0^-, 0^+)$: $LC[h'(0^+) - 0] = 1$ and $h(0^+) = 0$. So $A = 0$, $B\omega_0 = 1/LC \Rightarrow B = \omega_0$. **$h(t) = (1/\sqrt{(LC)})\sin(t/\sqrt{(LC)})\,u(t)$** — an undamped oscillation (marginally unstable).

(d) $v_R = x - v_C$, $RC\,v_C' + v_C = x$. For $x = \delta$: v_C jumps to $1/RC$ and decays: $h_C = (1/RC)e^{-t/RC}u(t)$. Hence **$h(t) = \delta(t) - (1/RC)e^{-t/RC}u(t)$** (the resistor passes the impulse instantly, then the capacitor's charge discharges through it).

(e) (i) $\delta(t)-\delta(t-2)$: memory, causal, stable; $2\text{rect}(t/2)$: memory, non-causal, stable; $2e^{-4t}u$: memory, causal, stable ($\int = 1$); $(1-e^{-4t})u$: memory, causal, unstable ($\int \to \infty$). (ii) $h = 2e^{-10t}u(t)$ (≈ 0 well before 5 s, so responses do not overlap); $y = \Sigma_{k=-2}^{2}\,2(-1)^k e^{-10(t-5k)}u(t-5k)$: five decaying spikes of alternating sign at $t = -10, -5, 0, 5, 10$.

Quick Notes — LTI and Convolution

Revision points for Lessons 6–10.

L6 · System Properties

- Linear: a x_1 + b x_2 → a y_1 + b y_2 (zero in → zero out). TI: x(t − t_0) → y(t − t_0).
- t outside x(·), x(2t), x(−t) ⇒ time-variant. x(t + a), x(−t) ⇒ non-causal.
- (1/3)[x(t) + x(t ± t_0)]: moving average — LTI, stable, non-causal.

L7 · Discrete Convolution

- c[n] = Σ a[k]b[n − k] = polynomial coefficients; length L + M − 1; Σy = ΣxΣh.
- {1,2,1,1} * {1,1,−1,2} = {1,3,2,2,4,1,2}.

L8 · Impulse Response

- x = Σx[k]δ[n − k] → y = Σx[k]h[n − k] (TI + homogeneity + additivity).
- Delay by 3: h = δ[n − 3]. Echo: h = u[n] or α^nu[n].

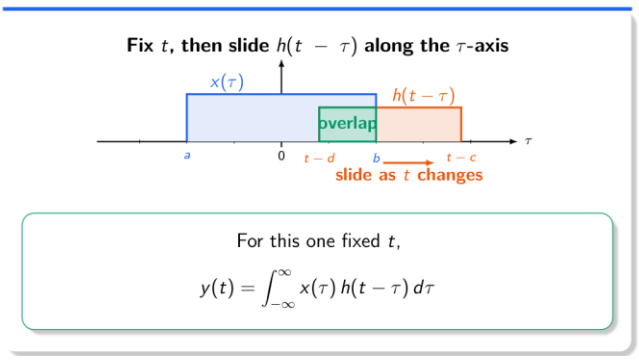
L9 · Convolution Practice

- α^nu * u = (1 − α^{n+1})/(1 − α) u; e^{-at}u * u = (1 − e^{-at})/a u.
- e^{-at}u * e^{-bt}u = (e^{-bt} − e^{-at})/(a − b) u; equal a ⇒ t e^{-at}u.
- Cascade: convolve; parallel: add; δ(t − a) * h = h(t − a).
- Rect on (a, b): y = H(t − a) − H(t − b). Interval table + continuity check.

L10 · Causality, Stability

- Causal ⇔ h = 0 for t < 0; stable ⇔ ∫|h| < ∞. Periodic h: neither.
- y′ + ay = bx ⇒ h = b e^{-at}u. LC ⇒ ω_0 sin ω_0t u(t). RC, v_R ⇒ δ − (1/RC)$e^{-t/RC}$u.

Key Diagrams to Reproduce



CT convolution: slide h(t − τ) along τ and integrate the overlap (Lec 4, sl. 7)